

LONGKOU, China

WMO: 547530

Lat: 37.6394N

Lon: 120.3359E

Elev: 16

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.7	21.1	3.0	6.4	26.9	5.3	7.2	27.9	19.6	27.6	18.0	27.7	5.7	180	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	91.2	76.6	89.0	75.5	86.9	74.9	80.5	88.0	79.2	85.8	78.0	84.1	8.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	147.4	85.6	77.2	141.7	84.1	76.1	136.2	82.9	44.3	88.2	42.9	85.9	41.6	84.2	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	16.5	14.7	15.2	96.2	2.7	3.0	13.2	98.3	11.6	100.1	10.1	101.8	8.1	103.9
			13.3	82.5	2.6	1.2	11.4	83.3	9.9	84.0	8.4	84.7	6.5	85.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.3	29.9	33.3	43.2	55.6	66.7	74.5	79.7	78.6	71.1	60.2	46.7	35.1
	DBStd	18.40	5.02	6.37	7.83	7.50	6.32	4.93	4.24	4.08	4.61	5.75	6.82	5.28
	HDD50	1970	622	468	241	27	0	0	0	0	0	4	144	464
	HDD65	4657	1087	888	676	293	57	2	0	0	7	169	550	928
	CDD50	4282	0	0	31	197	516	734	922	887	632	319	44	0
	CDD65	1494	0	0	1	12	108	286	457	422	188	20	0	0
	CDH74	12336	0	0	5	78	709	2166	4525	3831	972	50	0	0
	CDH80	3730	0	0	0	12	174	658	1587	1147	151	1	0	0
(14) Wind	WSAvg	7.3	6.9	7.2	8.2	8.7	8.1	7.5	6.7	6.1	6.1	6.8	7.5	7.5
(15) Precipitation	PrecAvg	24.3	0.3	0.4	0.5	1.3	1.7	2.8	6.7	5.4	2.5	1.4	0.9	0.4
	PrecMax	37.2	1.5	1.5	3.0	4.3	5.3	10.1	14.6	12.1	6.9	3.5	3.8	1.0
	PrecMin	15.8	0.0	0.0	0.0	0.1	0.0	0.2	1.6	1.1	0.1	0.1	0.0	0.0
	PrecStd	5.6	0.4	0.4	0.6	1.1	1.3	2.4	3.2	3.0	1.7	1.0	0.8	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	55.8	71.7	81.0	89.3	93.0	94.8	92.7	87.3	78.4	67.8	54.2
		MCWB	39.3	45.3	55.0	61.7	67.4	70.4	80.3	80.4	73.9	63.3	56.2	47.9
	2%	DB	43.4	50.3	64.4	75.8	84.5	89.2	91.9	89.9	83.6	75.2	63.6	49.1
		MCWB	37.2	41.1	50.8	59.0	65.1	70.4	78.4	78.6	70.6	62.8	53.8	43.0
	5%	DB	40.4	46.6	59.6	71.8	81.0	86.6	89.7	87.7	81.5	72.1	60.4	45.8
		MCWB	35.1	38.9	47.3	56.5	62.9	69.5	77.3	77.5	70.2	61.4	52.5	40.1
	10%	DB	37.7	43.3	55.2	68.0	77.9	83.8	87.3	85.6	79.2	69.3	57.5	43.1
		MCWB	32.9	36.9	44.8	54.5	61.7	68.9	76.6	76.4	68.8	60.2	50.3	37.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.9	46.8	55.7	64.6	70.1	76.9	82.3	82.4	77.4	67.8	59.0	48.6
		MCDB	45.2	53.5	68.7	76.7	84.5	86.6	91.5	90.4	82.6	74.2	63.6	52.1
	2%	WB	38.2	42.5	52.0	61.5	67.6	74.5	80.8	80.7	74.4	65.6	56.3	44.1
		MCDB	42.4	48.9	62.5	72.5	80.6	83.5	89.0	87.5	80.6	71.4	61.1	48.3
	5%	WB	35.7	39.9	48.6	58.4	65.7	72.5	79.6	79.4	72.5	63.7	53.6	41.0
		MCDB	39.7	45.2	57.6	69.1	76.8	81.2	87.0	85.4	78.5	69.8	59.5	44.5
	10%	WB	33.3	37.5	45.6	56.0	63.8	71.0	78.3	78.2	70.8	61.7	50.9	38.6
		MCDB	37.1	42.8	54.0	65.7	74.3	79.7	85.0	83.8	77.0	67.9	56.8	42.7
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.0	14.0	15.2	15.1	14.0	11.4	10.8	12.8	12.9	11.8	10.0
		MCDBR	15.3	17.2	20.0	20.5	19.8	18.3	14.4	12.7	14.8	16.2	16.2	14.2
		MCWBR	11.1	11.4	11.6	11.0	9.7	7.8	6.0	5.6	7.0	8.4	10.5	10.5
		MCDBR	13.4	15.1	18.2	19.0	17.0	14.1	12.3	11.0	12.0	13.5	14.0	12.7
	5% WB	MCWBR	10.6	10.9	11.5	11.1	9.6	7.0	6.1	5.6	6.9	8.6	10.7	10.5
(40) Clear-Sky Solar Irradiance	taub		0.490	0.546	0.589	0.623	0.651	0.753	0.739	0.671	0.592	0.614	0.562	0.505
	taud		1.812	1.724	1.667	1.653	1.639	1.513	1.591	1.703	1.799	1.674	1.736	1.810
	Ebn at Noon		200	207	215	218	215	194	195	205	212	186	177	184
	Edh at Noon		51	63	73	78	80	91	84	73	63	65	55	48
(44) All-Sky Solar Radiation	RadAvg		709	973	1391	1646	1822	1720	1514	1469	1344	1048	743	612
	RadStd		68	88	104	71	143	134	153	147	106	92	96	47

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page